THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 14: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A box contains 25 tickets numbered 1 to 25. One ticket is drawn at random. The probability that the number drawn is a perfect square is: 1

(a) $\frac{4}{25}$ **(b)** $\frac{1}{5}$ **(c)** $\frac{4}{5}$ **(d)** $\frac{6}{25}$

2. A lot of 90 ball pens contains 10 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

(a) $\frac{9}{10}$ **(b)** $\frac{8}{9}$ **(c)** $\frac{1}{9}$ **(d)** $\frac{79}{90}$

3. Two dice are thrown simultaneously. The probability that both dice show the same number (a doublet) is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{7}{36}$ **(c)** $\frac{5}{6}$ **(d)** $\frac{5}{36}$

4. A coin is tossed three times. The probability of getting all three faces the same is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{3}{8}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{8}$

5. A letter is chosen at random from the word 'EXAMINATION'. The probability that it is a consonant is: 1

(a) $\frac{5}{11}$ **(b)** $\frac{5}{12}$ **(c)** $\frac{6}{11}$ **(d)** $\frac{4}{11}$

6. A bag contains 3 red and 7 blue balls. 7 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{9}{17}$ **(b)** $\frac{7}{17}$ **(c)** $\frac{10}{17}$ **(d)** $\frac{11}{17}$

7. A box contains 30 tickets numbered 1 to 30. One ticket is drawn at random. The probability that the number drawn is a prime number is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{3}{10}$ **(d)** $\frac{11}{30}$

8. A lot of 75 ball pens contains 12 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

(a) $\frac{64}{75}$ **(b)** $\frac{62}{75}$ **(c)** $\frac{21}{25}$ **(d)** $\frac{4}{25}$

9. Two dice are thrown simultaneously. The probability that the product of the numbers is 12 is: 1

(a) $\frac{5}{36}$ **(b)** $\frac{8}{9}$ **(c)** $\frac{1}{9}$ **(d)** $\frac{1}{12}$

10. A coin is tossed three times. The probability of getting at most one tail is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{4}{9}$ **(c)** $\frac{3}{8}$ **(d)** $\frac{5}{8}$

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A red die and a blue die are rolled together. Find the probability that the numbers differ by 2. 2

12. A letter is chosen at random from the word 'SUCCESS'. Find the probability that it is the letter S. 2

13. A coin is tossed three times. Find the probability of getting heads on the first and last tosses. 2

14. A red die and a blue die are rolled together. Find the probability that both numbers are odd. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two dice are thrown simultaneously. Find the probability that the first number is greater than the second. 3

16. A box contains 40 tickets numbered 1 to 40. One ticket is drawn at random. Find the probability that the number drawn is a multiple of 4. 3

17. A bag contains 4 red and 8 blue balls. 4 more red balls are put into the bag and a ball is drawn at random. Find the probability that the ball drawn is red. 3

OR

Two dice are thrown simultaneously. Find the probability that the sum is 12. 3

18. A box contains 60 tickets numbered 1 to 60. One ticket is drawn at random. Find the probability that the number drawn is divisible by both 2 and 3. 3

OR

A bag contains 8 red and 6 blue balls. 2 more red balls are put into the bag and a ball is drawn at random. Find the probability that the ball drawn is red. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the numbers differ by 2,
(ii) both numbers are odd,
(iii) the sum is a prime number.

OR

A bag contains 3 red and 9 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{4}{7}$. Find the value of x. 5

20. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the product is a perfect square,
(ii) the sum is a multiple of 4,
(iii) the first number is greater than the second.

OR

A bag contains 6 red and 6 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{2}{3}$. Find the value of x. 5

END OF QUESTION PAPER — Total: 40 Marks